Early-pregnancy vitamin D and gene expression

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Methods

Data source and analytical sample

I analyzed the three instructor-provided CSV files derived from GEO GSE85307 (Mirzakhani et al. 2016). The original gene-expression substudy was nested within VDAART and included 47 women who later developed preeclampsia and 110 controls. The supplied data contain these 157 participants and 500 expression probes; this is a course subset, not the full transcriptome or the 881-person parent trial. Peripheral blood was collected in early pregnancy. GEO documents RMA-normalized log2 expression for the original array samples (National Center for Biotechnology Information, n.d.-c). I retained the supplied numeric scale, without further log transformation or re-normalization; the course files do not independently document any additional preprocessing.

I performed all data wrangling in R. I removed metadata prefixes before numeric conversion. I used unique sample identifiers to reorder expression columns to match covariate rows; original file orders differed. Using the instructor's linker file, I mapped all 500 supplied probes one-to-one to gene symbols. I found no missing values after conversion and excluded no participants. The initial and final analytical sample sizes were both 157.

Variables and visualization

I selected baseline 25-hydroxyvitamin D (25[OH]D, ng/mL) as my primary continuous covariate. I used subsequent pregnancy outcome and a vitamin D category (<30 versus >=30 ng/mL) as my categorical covariates. I derived the vitamin D category using the source study's threshold (Mirzakhani et al. 2016). This cut point is a study convention, not a universal clinical diagnostic criterion. I added maternal age and BMI to the summary table. I selected CYP24A1 as my primary gene because of its role in vitamin D metabolism (National Center for Biotechnology Information, n.d.-b). PTX3 and CXCR2 provide inflammatory and chemokine context (National Center for Biotechnology Information, n.d.-d, n.d.-a). I did not select genes on the basis of observed statistical significance.

I used histograms, unadjusted scatterplots, and grouped boxplots to describe gene expression. I added an ordinary least-squares line and a model-based 95% confidence band for the mean to the scatterplot. I interpreted slopes and correlations as descriptive and unadjusted. In Table 1, I summarized continuous variables using means, SDs, medians, and quartiles, and categorical variables using counts and within-column percentages.

For the heatmap, I used ten fixed genes available in the supplied data: CYP24A1, PTX3, CXCR2, JAK2, LYN, SOCS5, IL7, IL2, TGM2 and TIMP1. I included the latter seven genes as a fixed exploratory visualization panel rather than as a validated pathway signature. I standardized expression within each gene across participants. I clustered rows and columns using Euclidean distance and complete linkage and displayed the sample dendrogram. After clustering, I matched the annotation bars to samples by sample ID. I also used empirical cumulative distribution functions (ECDFs) to compare expression distributions for the three genes across four groups defined by pregnancy outcome and vitamin D category. At each x, the ECDF reports the proportion with expression at or below x.

Software and reproducibility

I used R (R Core Team 2026), ggplot2 (Wickham 2016), dplyr (Wickham et al. 2026), tidyr (Wickham et al. 2025), patchwork (Pedersen 2025) and rlang (Henry and Wickham 2026). I compiled the documents with knitr (Xie 2026) and rmarkdown (Allaire et al. 2026). I recorded the exact versions for this run below and in `results/sessionInfo.txt`. I used a fixed seed for point jitter only; it does not alter the observed measurements. I supplied input checksums and the probe mapping with the reproducible code.

Exact versions were R 4.6.1, ggplot2 4.0.3, dplyr 1.2.1, tidyr 1.3.2, patchwork 1.3.2, rlang 1.3.0, knitr 1.52, and rmarkdown 2.32.

The analysis code and submission history are available on GitHub: Project repository.

Results

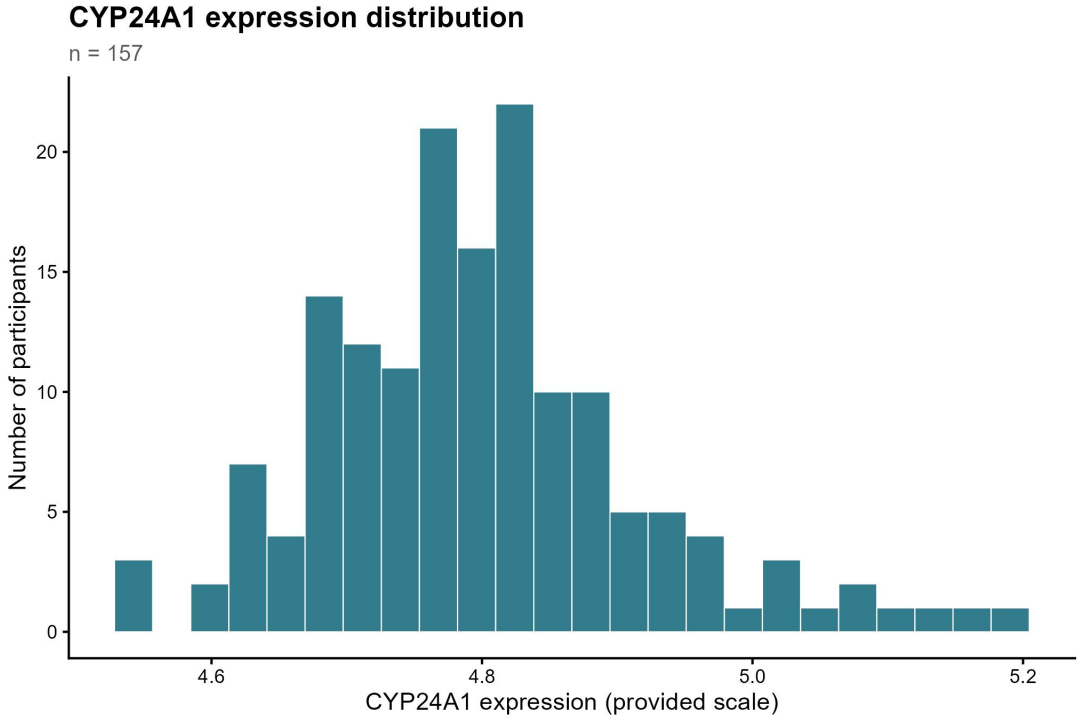
Sample characteristics

I included 110 controls and 47 participants who subsequently developed preeclampsia. Mean vitamin D was 24.37 ng/mL in controls and 19.72 ng/mL in cases; mean BMI was 27.99 and 31.16 kg/m2, respectively (Table 1). Vitamin D >=30 ng/mL occurred in 32 controls (29.1%) and 6 cases (12.8%). These are

descriptive contrasts within a selected case-control sample; the 47/157 fraction is not an estimate of population preeclampsia incidence.

Primary-gene distribution

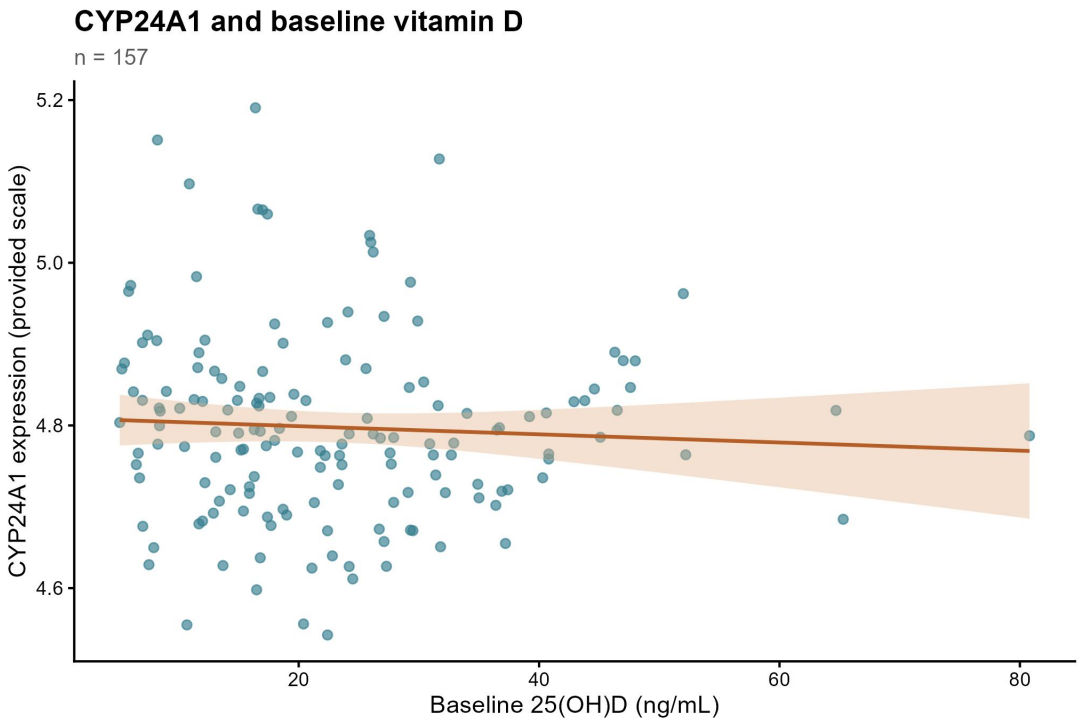
CYP24A1 had a median expression of 4.790 (Q1–Q3, 4.721–4.847) and a range of 4.542–5.191 on the supplied scale (Figure). Most values were near 4.8, with a modest upper tail. Normality was not established by the histogram.



Distribution of CYP24A1 expression among 157 participants. Each bar counts observations in one of 24 bins. No additional transformation or exclusion was applied.

Continuous vitamin D and CYP24A1

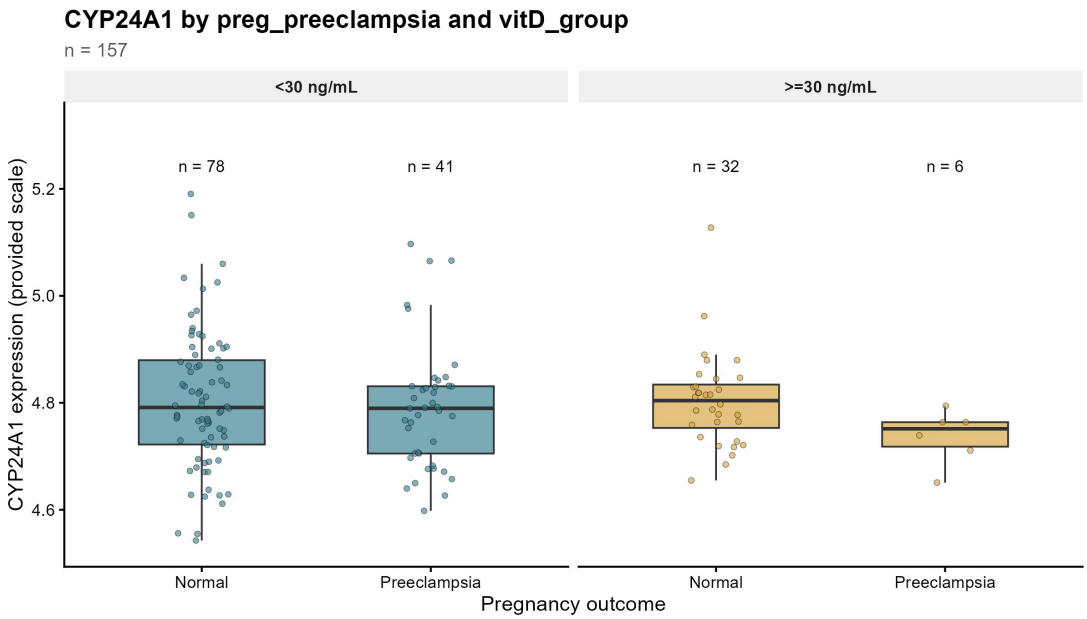
I obtained a descriptive Pearson correlation of -0.057. A 10-ng/mL higher vitamin D level corresponded to an unadjusted fitted mean expression difference of -0.005 (model-based 95% CI, -0.019 to 0.009). The fitted line was nearly flat (Figure). This provides little indication of a linear relationship in these data; it does not prove that all relationships are absent.



Baseline 25(OH)D and CYP24A1 expression (n=157). Points represent participants. The line is an unadjusted linear fit; shading is its model-based 95% confidence band for the mean, not an individual prediction interval.

CYP24A1 across both categorical covariates

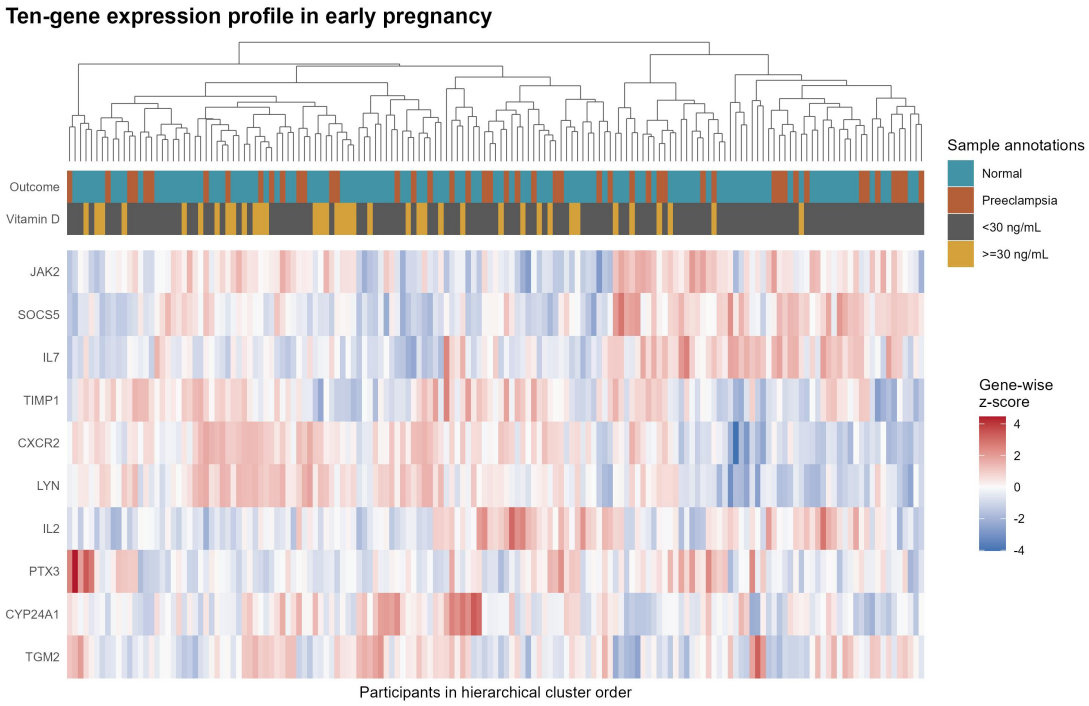
For vitamin D <30 ng/mL, median expression was 4.791 in controls (n=78) and 4.790 in cases (n=41). For vitamin D >=30 ng/mL, medians were 4.804 in controls (n=32) and 4.751 in cases (n=6). The distributions overlap (Figure). The smallest group's apparent difference is particularly uncertain; these comparisons do not establish an interaction.



CYP24A1 expression by pregnancy outcome and vitamin D category. Boxes show medians and interquartile ranges. Whiskers extend to observations within 1.5 times the IQR; every observation is overlaid. Group counts are displayed. Horizontal jitter is for visibility only.

Ten-gene expression patterns

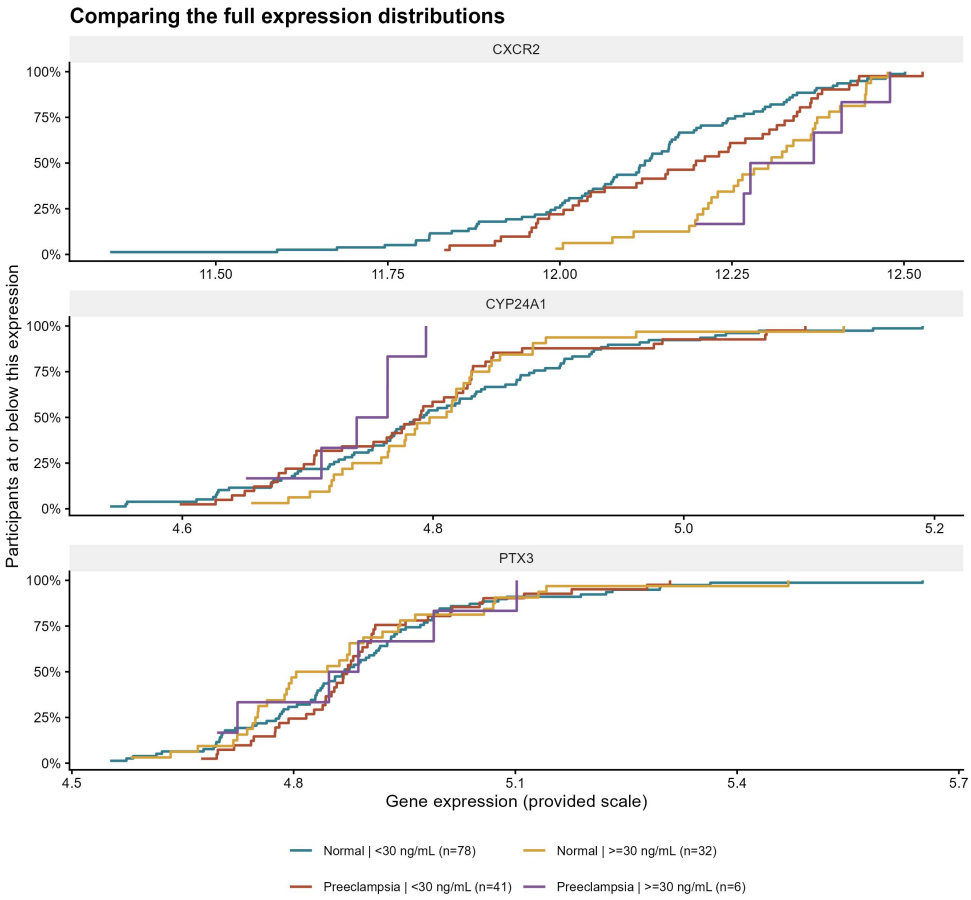
After standardizing expression within each gene, I observed relative expression patterns across participants, while the annotation bars show mixed pregnancy outcomes within the clustered sample ordering (Figure). Red indicates expression above that gene's sample mean; blue indicates expression below it. Colors cannot establish that one gene is more highly expressed than another. This exploratory clustering is neither a validated diagnostic classifier nor evidence of disease-specific molecular subtypes.



Ten-gene heatmap for 157 participants. Values are gene-wise z-scores. Both genes and participants are ordered by complete-linkage hierarchical clustering using Euclidean distance; the sample dendrogram is shown. Outcome and vitamin D annotation bars follow exactly the same sample ordering.

Additional visualization of distributional differences

In the ECDFs, I observed a rightward shift in CXCR2 expression for the higher-vitamin-D groups, while the CYP24A1 and PTX3 curves show more overlap (Figure). The CXCR2 descriptive correlation with continuous vitamin D was 0.346. The six-person subgroup has visibly larger ECDF steps because each participant accounts for about 16.7% of that group's distribution. These patterns motivate further investigation but are not adjusted causal estimates.



Empirical cumulative distributions of three genes by pregnancy outcome and vitamin D group. The y axis gives the percentage at or below each expression value. Each panel uses its own x-axis scale; group counts appear in the legend. ECDFs summarize the full observed distribution without a bin width or smoothing bandwidth.

Interpretation and limitations

I conducted descriptive, observational analyses despite the parent study being a randomized supplementation trial: baseline vitamin D and subsequent disease status were not randomized exposures in this analysis. Selection into the nested case-control sample limits population interpretations. I did not adjust for maternal BMI, age, blood-cell composition, gestational timing, matching factors, or array batch. The source study used more extensive processing and analysis than this visualization project; I do not claim to reproduce that analysis.

The provided matrix contains only 500 probes. Its selection mechanism and any additional course-specific preprocessing are not documented in the supplied files. A gene absent from this matrix cannot be evaluated here. Some recorded BMI

values repeat and one collection time exceeds 18 weeks; without evidence of an error, I retained these values. The high-vitamin-D case subgroup is small. Because categorizing vitamin D loses information, I also retained the continuous analysis. I make no confirmatory claims about statistical significance across multiple genes.

References

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